

tf.reshape

 View source on GitHub (https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L63-L201)

Reshapes a tensor.

```
tf.reshape(  
    tensor, shape, name=None  
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Introduction to Tensors (https://www.tensorflow.org/guide/tensor) • Advanced automatic differentiation (https://www.tensorflow.org/guide/advanced_autodiff) • DTensor concepts (https://www.tensorflow.org/guide/dtensor_overview) • Multilayer perceptrons for digit recognition with Core APIs (https://www.tensorflow.org/guide/core/mlp_core) • Distributed training with Core APIs and DTensor (https://www.tensorflow.org/guide/core/distribution)	• Learned data compression (https://www.tensorflow.org/tutorials/generative/data_compression) • Adversarial example using FGSM (https://www.tensorflow.org/tutorials/generative/adversarial_fgsm) • Convolutional Variational Autoencoder (https://www.tensorflow.org/tutorials/generative/cvae) • Load CSV data (https://www.tensorflow.org/tutorials/load_data/csv) • Load text (https://www.tensorflow.org/tutorials/load_data/text)

Given `tensor`, this operation returns a new `tf.Tensor` (https://www.tensorflow.org/api_docs/python/tf/Tensor) that has the same values as `tensor` in the same order, except with a new shape given by `shape`.

```
>>> t1 = [[1, 2, 3],  
...      [4, 5, 6]]  
>>> print(tf.shape(t1).numpy())  
[2 3]  
>>> t2 = tf.reshape(t1, [6])  
>>> t2  
<tf.Tensor: shape=(6,), dtype=int32,  
  numpy=array([1, 2, 3, 4, 5, 6], dtype=int32)>  
>>> tf.reshape(t2, [3, 2])  
<tf.Tensor: shape=(3, 2), dtype=int32, numpy=  
  array([[1, 2],  
         [3, 4],  
         [5, 6]], dtype=int32)>
```

The `tf.reshape` (https://www.tensorflow.org/api_docs/python/tf/reshape) does not change the order of or the total number of elements in the tensor, and so it can reuse the underlying data buffer. This makes it a fast operation independent of how big of a tensor it is operating on.

```
>>> tf.reshape([1, 2, 3], [2, 2])  
Traceback (most recent call last):  
...  
InvalidArgumentError: Input to reshape is a tensor with 3 values, but the  
requested shape has 4
```

To instead reorder the data to rearrange the dimensions of a tensor, see `tf.transpose` (https://www.tensorflow.org/api_docs/python/tf/transpose).

```
>>> t = [[1, 2, 3],  
...      [4, 5, 6]]  
>>> tf.reshape(t, [3, 2]).numpy()  
array([[1, 2],  
       [4, 5],  
       [3, 6]])
```

```

        [3, 4],
        [5, 6]], dtype=int32)
>>> tf.transpose(t, perm=[1, 0]).numpy()
array([[1, 4],
       [2, 5],
       [3, 6]], dtype=int32)

```

If one component of `shape` is the special value -1, the size of that dimension is computed so that the total size remains constant. In particular, a `shape` of `[-1]` flattens into 1-D. At most one component of `shape` can be -1.

```

>>> t = [[1, 2, 3],
...      [4, 5, 6]]
>>> tf.reshape(t, [-1])
<tf.Tensor: shape=(6,), dtype=int32,
  numpy=array([1, 2, 3, 4, 5, 6], dtype=int32)>
>>> tf.reshape(t, [3, -1])
<tf.Tensor: shape=(3, 2), dtype=int32, numpy=
  array([[1, 2],
        [3, 4],
        [5, 6]], dtype=int32)>
>>> tf.reshape(t, [-1, 2])
<tf.Tensor: shape=(3, 2), dtype=int32, numpy=
  array([[1, 2],
        [3, 4],
        [5, 6]], dtype=int32)>

```

`tf.reshape(t, [])` reshapes a tensor `t` with one element to a scalar.

```

>>> tf.reshape([7], []).numpy()
7

```

More examples:

```

>>> t = [1, 2, 3, 4, 5, 6, 7, 8, 9]
>>> print(tf.shape(t).numpy())
[9]
>>> tf.reshape(t, [3, 3])
<tf.Tensor: shape=(3, 3), dtype=int32, numpy=
  array([[1, 2, 3],
        [4, 5, 6],
        [7, 8, 9]], dtype=int32)>

```

```

>>> t = [[[1, 1], [2, 2]],
...      [[3, 3], [4, 4]]]
>>> print(tf.shape(t).numpy())
[2 2 2]
>>> tf.reshape(t, [2, 4])
<tf.Tensor: shape=(2, 4), dtype=int32, numpy=
  array([[1, 1, 2, 2],
        [3, 3, 4, 4]], dtype=int32)>

```

```

>>> t = [[[1, 1, 1],
...      [2, 2, 2]],
...      [[3, 3, 3],
...      [4, 4, 4]],
...      [[5, 5, 5],
...      [6, 6, 6]]]
>>> print(tf.shape(t).numpy())
[3 2 3]

```

```
>>> # Pass '[-1]' to flatten 't'.
>>> tf.reshape(t, [-1])
<tf.Tensor: shape=(18,), dtype=int32,
  numpy=array([1, 1, 1, 2, 2, 2, 3, 3, 3, 4, 4, 4, 5, 5, 5, 6, 6, 6]),
  dtype=int32)>
>>> # -- Using -1 to infer the shape --
>>> # Here -1 is inferred to be 9:
>>> tf.reshape(t, [2, -1])
<tf.Tensor: shape=(2, 9), dtype=int32, numpy=
  array([[1, 1, 1, 2, 2, 2, 3, 3, 3],
        [4, 4, 4, 5, 5, 5, 6, 6, 6]], dtype=int32)>
>>> # -1 is inferred to be 2:
>>> tf.reshape(t, [-1, 9])
<tf.Tensor: shape=(2, 9), dtype=int32, numpy=
  array([[1, 1, 1, 2, 2, 2, 3, 3, 3],
        [4, 4, 4, 5, 5, 5, 6, 6, 6]], dtype=int32)>
>>> # -1 is inferred to be 3:
>>> tf.reshape(t, [ 2, -1, 3])
<tf.Tensor: shape=(2, 3, 3), dtype=int32, numpy=
  array([[[1, 1, 1],
         [2, 2, 2],
         [3, 3, 3]],
        [[4, 4, 4],
         [5, 5, 5],
         [6, 6, 6]]], dtype=int32)>
```

Args	
tensor	A Tensor .
shape	A Tensor . Must be one of the following types: int32 , int64 . Defines the shape of the output tensor.
name	Optional string. A name for the operation.
Returns	
A Tensor . Has the same type as tensor .	

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